

Data science in chemical engineering

Installation guide

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1 Base Installation

Many tools and development environments can run Python scripts on your machine or online. My recommendation is **Visual Studio Code (VSC)** from Microsoft. If you prefer to use some other alternatives, I will provide a list of other recommendations down below. VSC installation is very straightforward.

For MacOS, you can follow this tutorial:

<https://www.youtube.com/watch?v=QmFOI8GYz6E&t=1s>

For Windows, you can follow this tutorial:

<https://www.youtube.com/watch?v=kCMJxOSiWrg>

If you use any Linux distro, I consider you skilled enough to have either done this already or be able to do it yourself :).

Try running this simple code that prints the value of the square root of two to check that everything is installed correctly:

```
1 import math
2
3 a = math.sqrt(2)
4 print(a)
```

2 Installing packages

Python packages are collections of pre-written code (modules) that add functionality to your Python projects. Think of them as add-ons or extensions that let you do more without writing everything from scratch. For example:

- **matplotlib** helps you create graphs and visualizations
- **numpy** provides advanced mathematical operations
- **scikit-learn** machine learning library with datasets and tools

Think of this as avoiding "reinventing the wheel" for common tasks.

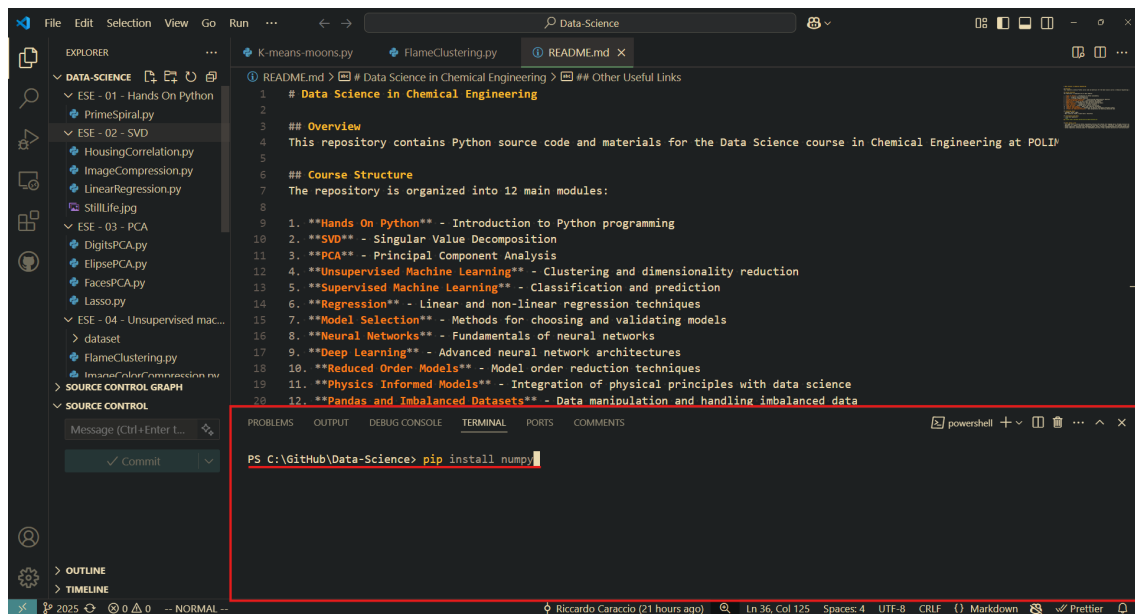
To install a package, simply go into the VSC terminal and type:

```
pip install package_name
```

Try installing the **numpy** and **matplotlib** packages. See the example below for more details.

If the command does not work you can also try:

```
python -m pip install package_name
# or
python3 -m pip install package_name
```



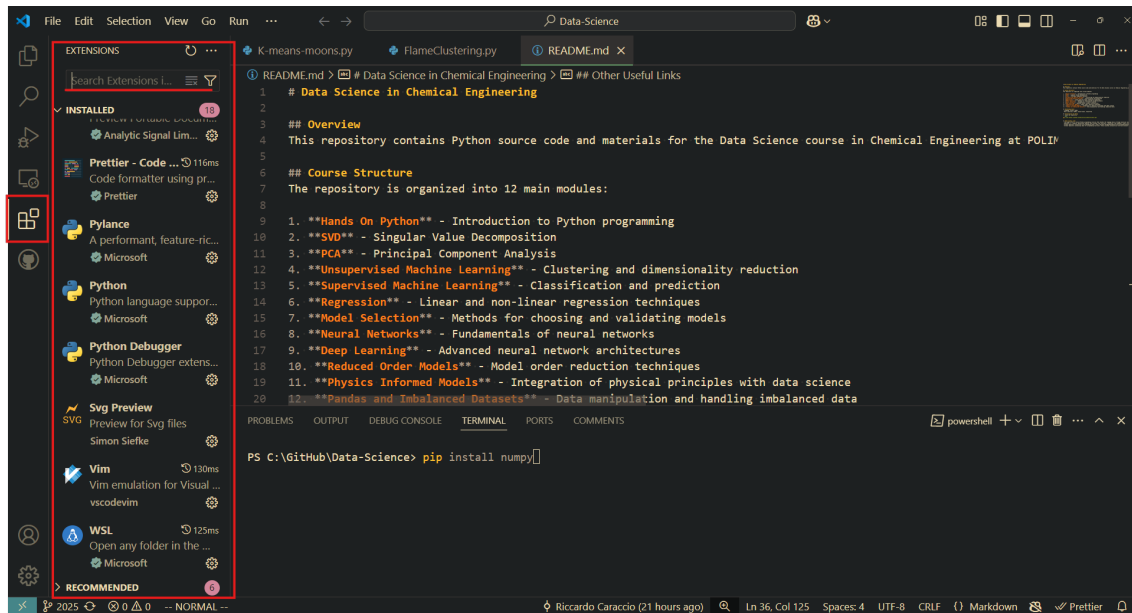
Let's run a simple code to check that you have done everything correctly.

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 x = np.linspace(0, 10, 100)
5 y = np.sin(x) + np.random.normal(0, 0.2, 100)
6
7 plt.plot(x, y, label='Random Data')
8 plt.xlabel('X-axis')
9 plt.ylabel('Y-axis')
10 plt.title('Numpy and Matplotlib Example')
11 plt.legend()
12 plt.show()
```

Try to understand what this code is doing!

3 Extensions

Visual Studio Code extensions are add-ons that enhance the functionality of your code editor. They're like apps you can install to make coding easier and more productive. You already installed the basic extension for the support of the Python language during the installation procedure. Remember that you can install them from the corresponding tab, as shown below:



For a better coding experience, I suggest that you check out **Pylance** and **Python Debugger**. Many more are available in the environment; various let you personalize the appearance of the code (Search 'Theme' for this kind of stuff) and others add additional help to support you in writing code. You decide!

4 GitHub page of the course

Lastly, I would like to mention that at the following link:

<https://github.com/Riccaraccio/Data-Science>

you will find the GitHub page of the course, containing all the scripts that we will see during the practicals. I will explain more in detail during class this powerful resource.

This is it for the installation guide!

5 Alternatives to Visual Studio Code

Here are several popular alternatives to Visual Studio Code for Python development:

- **PyCharm**
 - Professional IDE by JetBrains
 - Features: Advanced debugging, integrated testing, smart code navigation
 - Two versions: Free (Community) and paid (Professional)
 - Great for large projects
 - Steeper learning curve but very powerful
- **Google Colab**
 - Free, cloud-based Jupyter notebook environment
 - No setup required - runs in browser
 - Free GPU/TPU access
 - Easy sharing and collaboration
 - Direct integration with Google Drive
- **Jupyter Notebook/JupyterLab**
 - Perfect for data science and analysis
 - Interactive coding with cells
 - Great for visualizations and documentation
 - Web-based interface
 - Built-in markdown support
- **Spyder**
 - Specifically designed for scientific Python
 - Similar to MATLAB interface
 - Integrated with scientific libraries
 - Great for data analysis
 - Free and open source
- **Sublime Text**
 - Very fast and lightweight
 - Highly customizable
 - Good for multiple languages
 - Free to try, paid for long-term use
 - Popular for its speed and minimalist design
- **Atom**
 - Free and open source
 - Highly customizable
 - Good GitHub integration
 - Large community of package developers
 - Similar feel to VSCode
- **IDLE**
 - Comes bundled with Python
 - Very basic but good for beginners
 - No setup required
 - Limited features
 - Good for learning Python basics